

Total No. of Questions: 8]

SEAT No. :

PA-1296

[5925]-327

[Total No. of Pages : 4

S.E. (Artificial Intelligence and Data Science)

STATISTICS

(2019 Pattern) (Semester-IV) (217528)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Q.1 or Q.2 Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Calculate: [10]

- i) Quartile deviation (Q.D.),
- ii) Mean Deviation (M.D.) from mean, for the following data:

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of students	6	5	8	15	7	6	8

- b) The variables X and Y are connected by the equation $aX + bY + c = 0$. Show that the correlation between them is -1 if the signs of a and b are alike and +1 if they are different. [8]

OR

Q2) a) An analysis of monthly wages paid to the workers of two firms A and B belonging to the same industry give the following results: [10]

	Firm A	Firm B
Number of workers	500	600
Average daily wage	Rs. 186.00	Rs. 175.00
Variance of distribution of wages	81	100

- i) Which firm, A or B, has a larger wage bill?
- ii) In which firm, A or B, is there greater variability in individual wages?
- iii) Calculate (a) the average daily wage, and (b) the variance of the distribution of wages of all the workers in the firm is A and B taken together.

P.T.O.

- b) In a partially destroyed laboratory, record of an analysis of correlation data, the following results only are legible: [8]

Variance of $X=9$. Regression equations: $8X-10Y+66=0$, $40X-18Y=214$. what are:

- the mean values X and Y ,
- the correlation coefficient between X and Y ,
- the standard deviation of Y ?

- Q3) a)** A Dice is thrown 10 times. If getting an odd number is a success. What is the probability of getting [5]

- 8 successes
- at least 6 success?

- b) Fit Poisson's distribution to following data and calculate theoretical frequencies. [6]

x	0	1	2	3	4
f	122	60	15	2	1

- c) In a Sample of 1000 cases the means of a certain test is 14 and standard deviation is 2.5 assuming the distribution to be normal find [6]

- How many students scored between 12 & 15.
- How many scored below 8.

[Given: $A(z = 0.8) = 0.2881$, $A(z = 0.4) = 0.1554$, $A(z = 2.4) = 0.4918$]

OR

- Q4) a)** A Random variable X with following probability distribution [5]

X	1	2	3	4	5	6	7
$P(X)$	k	$2k$	$3k$	k^2	k^2+k	$2k^2$	$4k^2$

Find.

- k
- $P(x > 5)$
- $P(1 \leq x \leq 5)$

- b) In a continuous distribution density function [6]

$$f(x) = kx^2(1-x^3), 0 \leq x \leq 1.$$

Find the value of

- k
- Mean
- Variance

- c) MNC company conducted 1000 candidates' aptitude test. The average score is 45 and the standard deviation of score is 25. Assuming normal distribution for the result. [6]

Find

- The number of candidate whose score exceed 60.
- The number of candidates whose score lies between 30 & 60.

[Given: $A(z = 0.6) = 0.2257$]

- Q5) a)** In an experiment of pea breeding, the following frequencies of seeds were obtained. [6]

Round and green	Wrinkle and green	Round and yellow	wrinkle and yellow	Total
222	120	32	150	524

Theory predicts that the frequencies should be in the proportion 8:2:2:1. Examine the correspondence between theory and experiment. Given $\chi^2_{(0.05, 3)} = 7.815$

- The average marks in mathematics of a sample of 100 students was 51 with standard deviation of 6 marks. Could this have a random sample from the population with average marks 50? Given Z_{α} at 5% level of significance = 1.96 [6]
- A random sample of 16 newcomers gave a mean of 1.67 m and standard deviation of 0.16 m. Is the mean height of newcomers significantly different from the mean height of students' population of the previous year? Given $t_{0.05, 15} = 2.13$ [6]

OR

- Q6) a)** Following table shows number of books issued on the various days of week from a certain library. At 5% level of significance test the null hypothesis that number of books issued is independent of the day. [6]

Day	Mon.	Tue.	Wed.	Thurs.	Fri.	Sat.
No. of books issued	120	130	110	115	135	110

Given: Chi-square value at 5% level of significance for degrees of freedom 5 is 11.071.

- A random sample of 900 members has mean 3.4 cms. Can it be reasonably regarded as a sample from a large population of mean 3.2 cms and standard deviation 2.3 cms. [6]

- c) Find the F-statistics form the following data: [6]

Sample	size (n)	Total observation $\sum x$	Sum of squares of observations
1	8	9.6	61.52
2	11	16.5	73.26

- Q7) a) State & Prove Neyman-Pearson Fundamental Lemma. [9]
 b) Given the frequency function [8]

$$f(x, \theta) = \frac{1}{\theta}; 0 \leq x \leq \theta$$

$$= 0; \text{elsewhere}$$

And that you are testing the null hypothesis $H_0 : \theta = 1$ vs $\theta = 2$ by means of a single observed value of x . what would be the size of Type I and Type II error. If you choose the interval

- i) $0.5 \leq x$
 ii) $1 \leq x \leq 1.5$

Also obtain the power function of the test.

OR

- Q8) a) Write short notes on [8]
 i) Most powerful test
 ii) Uniformly most powerful test
 iii) Advantages and disadvantages of non-parametric tests
 iv) Level of significance
 b) Explain in detail about test for the Equality of means of serveral normal populations. [9]

